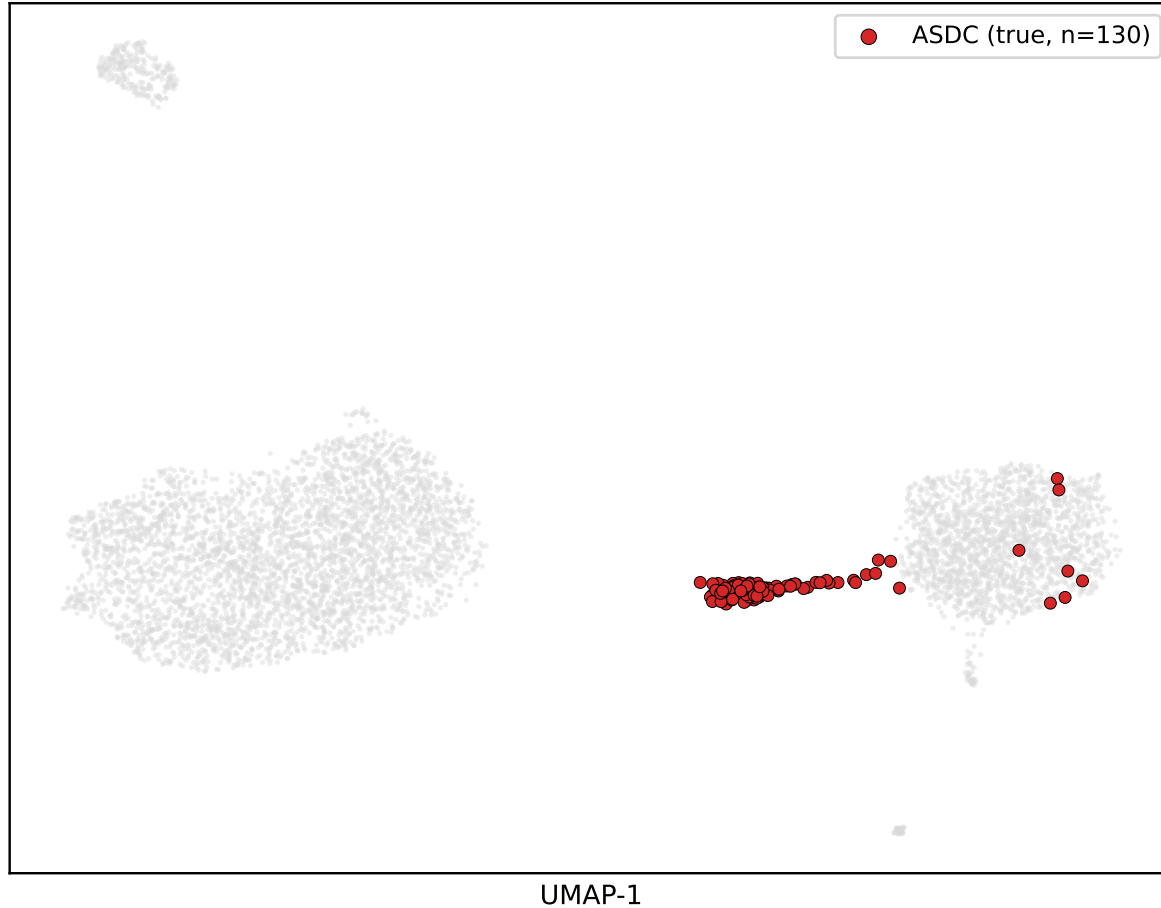


Why scRareRefine's gain shrinks at low separability

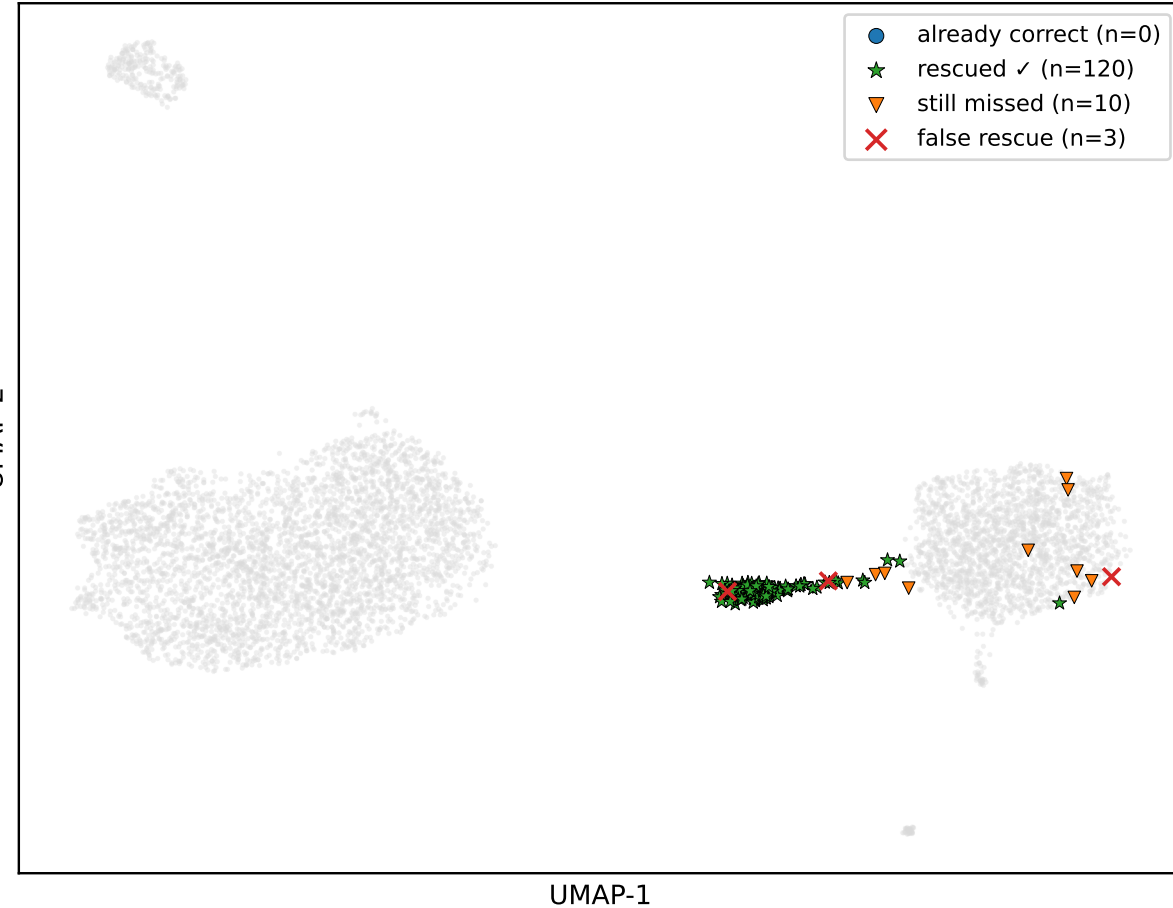
high-sep (immune_dc, sep=2.4): isolated rare cluster → scANVI misses all → large rescue (TP=120, FP=3)

low-sep (pancreas_baron, sep=1.4): rare cluster abuts majority → scANVI already finds 67% → limited rescue (TP=6, missed=22)

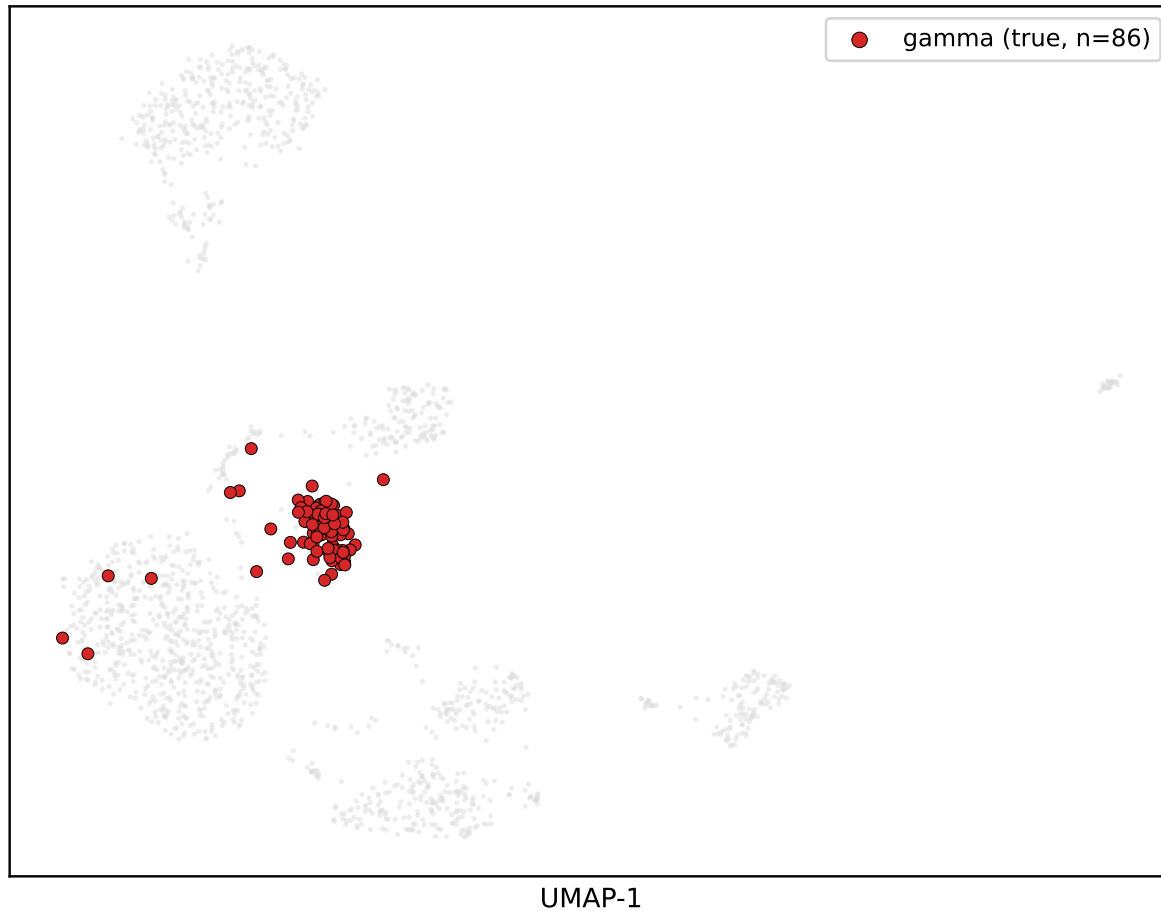
(a) immune_dc — ground truth (sep=2.39)



(b) rescue outcome — recall 0.00→0.92, prec 0.98



(c) pancreas_baron — ground truth (sep=1.40)



(d) rescue outcome — recall 0.67→0.74, prec 0.96

